

Categorical Library in Anders

Groupoid Infinity

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Abstract

This article describes the design and implementation of the categorical library in the Anders proof assistant. We focus on incorporating the design principles from Jacques Carette et al (2021) formalization in Agda, specifically regarding definitional duality and the use of cubical foundations for univalent categories.

Contents

1	Introduction	2
2	Definitional Duality	2
3	Cubical Foundations	2
4	Core Library Structure	2
5	Natural Transformations and Isomorphisms	3
6	Adjunctions	3
7	Equivalences of Categories	3
8	The Category of Categories	4
9	The Yoneda Lemma	4
10	Conclusion	5

1 Introduction

The Anders categorical library is designed to be a minimal yet powerful implementation of category theory in a pure Martin-Löf Type Theory (MLTT) with cubical extensions. The library prioritizes proof-relevance, univalence, and duality-friendliness.

2 Definitional Duality

A key design choice, following [1], is the use of symmetric laws for associativity and identity. By including both `assoc` and `assoc-inv`, as well as `id-id` (the neutral identity law), we ensure that the opposite category operation $(\cdot)^{\text{op}}$ is a definitional involution:

$$(\mathcal{C}^{\text{op}})^{\text{op}} \equiv \mathcal{C} \tag{1}$$

This allows for "duality for free," where any theorem proved for a category $\mathcal{C}$ can be immediately applied to its dual without explicit transport along an isomorphism.

3 Cubical Foundations

Unlike the Agda formalization which often relies on setoids to maintain compatibility with different Agda modes, Anders is natively cubical. This allows us to use Path types for equality between morphisms, which are proof-relevant by default. A univalent category in Anders is a precategory where the map from paths between objects to isomorphisms is an equivalence.

4 Core Library Structure

The core library consists of the following modules:

- `category.anders`: Basic definitions of precategories and categories.
- `functor.anders`: Functors and their compositions.
- `natural.anders`: Natural transformations and isomorphisms.
- `adjunction.anders`: Adjunctions between categories.
- `yoneda.anders`: Yoneda lemma.

5 Natural Transformations and Isomorphisms

Natural transformations are implemented as Sigma types consisting of the component map η and a naturality condition. We have formalized the definitional duality for natural transformations: if $\alpha : F \Rightarrow G$ is a natural transformation between functors $F, G : \mathcal{C} \rightarrow \mathcal{D}$, then α also defines a natural transformation $\alpha^{\text{op}} : G^{\text{op}} \Rightarrow F^{\text{op}}$ between the dual functors. In Anders, this duality is definitional, meaning $(\alpha^{\text{op}})^{\text{op}} \equiv \alpha$.

6 Adjunctions

Adjunctions $L \dashv R$ between categories $\mathcal{C}$ and $\mathcal{D}$ are defined via the unit-counit formulation:

- A unit natural transformation $\eta : \text{id}_{\mathcal{C}} \Rightarrow R \circ L$.
- A counit natural transformation $\varepsilon : L \circ R \Rightarrow \text{id}_{\mathcal{D}}$.
- Two triangle identities (zig and zag).

Due to the current limitations of the Anders compiler regarding named projections within nested Sigma types, the implementation uses numeric indices (e.g., `u.1` for η) to ensure stable typechecking.

7 Equivalences of Categories

A strong equivalence between two categories $\mathcal{C}$ and $\mathcal{D}$ is a pair of functors $F : \mathcal{C} \rightarrow \mathcal{D}$ and $G : \mathcal{D} \rightarrow \mathcal{C}$ equipped with natural isomorphisms:

- $\eta : \text{id}_{\mathcal{C}} \cong G \circ F$
- $\varepsilon : F \circ G \cong \text{id}_{\mathcal{D}}$

We have implemented `opEquivalence`, which shows that an equivalence $e : \mathcal{C} \cong \mathcal{D}$ induces an equivalence $e^{\text{op}} : \mathcal{C}^{\text{op}} \cong \mathcal{D}^{\text{op}}$. To ensure definitional duality, we utilize `symNatIso` to adjust the directions of the unit and counit in the dualized structure.

8 The Category of Categories

The category of small precategories, denoted $\mathcal{CAT}$, is formalized as a large category (of type U_2). The objects are precategories of type U_1 , and morphisms are functors of type U . While $\mathcal{CAT}$ is naturally a 2-category, we provide its 1-categorical structure by specifying identity functors and functor composition. The operation of taking the opposite category is currently provided as a definitional involution on the objects of $\mathcal{CAT}$.

9 The Yoneda Lemma

The Yoneda Lemma is a fundamental result in category theory that relates natural transformations between a hom-functor and an arbitrary functor to the values of that functor. For a small precategory $\mathcal{C}$, an object $A \in \mathcal{C}$, and a functor $F : \mathcal{C} \rightarrow \mathcal{SET}$, the lemma states:

$$\text{Nat}(\mathcal{C}(A, -), F) \cong F(A) \quad (2)$$

Our implementation addresses several technical challenges inherent in formalizing the Yoneda Lemma:

- **Large Target Category:** Standard functors in our library are restricted to small precategories. Since $\mathcal{SET}$ is a large category, we manually define `catfunSET` and `nattransSET` using Sigma types that accommodate objects from higher universes.
- **Cubical Isomorphism:** The isomorphism is established via the projection of `YonedaMap` and its inverse `YonedaInv`. The computability property of beta reduction `Yoneda-β` holds definitionally, while uniqueness property of isomorphism inverse `Yoneda-η` is proved by establishing a path between natural transformations.
- **Identification of Transformations:** To prove that two natural transformations are equal, we utilize `nattransSET-isProp`, which shows that the naturality condition for functors into $\mathcal{SET}$ is a mere proposition. This allows us to use `lemSig'` from the foundation library to establish the equality by only checking the component map η .

10 Conclusion

The Anders categorical library continues to expand by incorporating higher-level structures while maintaining strict adherence to definitional duality and cubical principles. The use of symmetric laws and explicit path-based proofs allows for a clean and performant formalization that avoids the overhead of traditional setoid-based approaches.

References

- [1] Jason Z. S. Hu and Jacques Carette. *Formalizing Category Theory in Agda*. CPP 2021.